Assignment (CN) → 02

Ques(1):- What are the different Protocol you observe at following layers of the Protocol stack?

(A) Application Layer.
(B) Transport Layer
(C) Network Layer

Ans→ (A) **Application layer Protocols :-**

- HTTP
- HTTPS (TLS)
- DNS
- mDNS
- QUIC

(B) **Transport layer Protocol used :-**

- TCP
- UDP

(C) **Network Layer Protocol used :-**

- IPV_4
- IPV_6
- ICMP (IPV_4)
- $ICMPV_6$
- ARP

Question(2):- What is the total amount of data being received for following two cases?

(A) When you access http://neverssl.com

Ans→ 3,182 Byte.

(B) When you access http://www.cornell.edu

Ans→ 76,447 Byte.

Ques(3):- Answer when you access the site http://neversell.com

(A) How many HTTP GET request do you observed list down the GET Request?

Ans→ (i) GET/online HTTP/1.1 (sent twice - bcoz of delay)
(ii) GET/online/HTTP/1.1
(iii) GET/favicon.ico HTTP/1.1

(B) What version of HTTP the surver is running?

Ans→ HTTP/1.1

(C) For Each of the HTTP GET request, findout:

(i) The total number of TCP segment being received.

Ans→

Get request	TCP request received
(i) GET/online	1 segment
(ii) GET/online/	2 segment
(iii) GET/favicon.ico	1 segment

(ii) The total amount of data being received in the corresponing HTTP response.

Ans→

Get request	Data received
(i) GET/online	577 Byte.
(ii) GET/online/	1,519 Byte
(iii) GET/favicon.ico	416 Byte.

(D) Identify TCP three way handshake used to establish the connection with the http://neverssl.com, specify SYN, SYN-ACK, Ack Packets and write the sequence number for each of the three access.

Ans→

Packet	Plags	sequence Number
1	SYN	Seq = 173939723
2	SYN-ACK	Seq = 1979181040 ACK = 173939724
3	ACK	Seq = 173939724 ACK = 1979181041